

Functional Requirements Document (FRD)

School Management System (SMS)

Project Name: School Management System

Version: 1.0

Prepared By: Jennifer Francis-Nkporwi (Business Analyst)

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1. Document Purpose

This Functional Requirements Document (FRD) defines the functional requirements for a School Management System (SMS). The system will automate and manage the administrative, academic, and financial operations of a school.

The FRD serves as a guide for stakeholders, developers, testers, and project managers throughout the software development lifecycle.

2. Project Overview

The School Management System is a web-based application designed to streamline school operations, including:

- Student registration and enrolment
- Attendance tracking
- Result Management
- Fee management
- Communication
- Reporting

3. Objectives

The system aims to:

- Reduce manual administrative work
- Improve accuracy and efficiency
- Provide centralized data management
- Enable real-time access to information
- Improve communication between school stakeholders
- Generate reports for decision-making

4. Stakeholders

Stakeholder	Role
School Administrator	Oversees all operations
Teachers	Manage attendance, grades, assignments
Parents	Monitor student progress
Finance Officer	Manage fees and payments
System Administrator	Manage system settings and security

5. Scope

In Scope

1. Student Registration
2. Class Assignment
3. Attendance Management
4. Fee Management
5. Result Management
6. Communication and Notifications
7. Reporting Dashboard

Out of Scope

- Payroll Management
- Timetable Scheduling
- Learning Management System (LMS)
- Online Classes
- Library Management
- Transport Management

6. Stakeholders

Stakeholder	Role
Admin	Manages student registration, class assignment, result approval, reporting
Teacher	Records attendance and enters scores
Finance Officer	Records payments and generates receipts
Parent	Views results and receives notifications

7. Functional Requirements

7.1 Student Registration Management

Module Description

Supports the registration of new students and assignment to classes.

Requirement ID	Functional Requirement
FR-001	The system shall allow the Admin to register a new student.
FR-002	The system shall capture student biodata and parent/guardian details.
FR-003	The system shall generate a unique Student ID automatically.
FR-004	The system shall prevent duplicate student records.
FR-005	The system shall allow the Admin to assign a registered student to a class for a selected academic session.

Requirement ID	Functional Requirement
FR-006	The system shall maintain one active class assignment per student per academic session.

7.2 Attendance Management

Module Description

Allows teachers to record and monitor daily attendance.

Requirement ID	Functional Requirement
FR-007	The system shall allow Teachers to record daily attendance by class and date.
FR-008	The system shall support attendance statuses: Present, Absent, and Late.
FR-009	The system shall prevent duplicate attendance submissions for the same class and date.
FR-010	The system shall store attendance records for reporting purposes.
FR-011	The system shall send notifications to parents when a student is marked absent.

7.3 Fee Management

Module Description

Supports recording of fee payments and generation of receipts.

Requirement ID	Functional Requirement
FR-012	The system shall allow the Finance Officer to record student fee payments.
FR-013	The system shall capture payment amount, payment date, and payment method.
FR-014	The system shall verify bank transfer payments before posting them.
FR-015	The system shall update the student outstanding balance automatically after payment is recorded.
FR-016	The system shall generate a receipt with a unique receipt number.

Requirement ID	Functional Requirement
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FR-017	The system shall allow parents to view outstanding balances.
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7.4 Result Management

Module Description

Supports score entry, grade calculation, result approval, and publication.

Requirement ID	Functional Requirement
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FR-018	The system shall allow Teachers to enter assessment and examination scores.
FR-019	The system shall validate score ranges based on configured limits.
FR-020	The system shall calculate totals, averages, and grades automatically.
FR-021	The system shall allow the Admin to review and approve submitted results.
FR-022	The system shall allow approved results to be published.
FR-023	The system shall allow parents to view published results online.

7.5 Communication and Notifications

Module Description

Provides automated notifications to parents.

Requirement ID	Functional Requirement
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FR-024	The system shall send notifications for student absences.
FR-025	The system shall send notifications when results are published.
FR-026	The system shall send fee-related reminders and alerts.
FR-027	The system shall maintain a history of notifications sent.

7.6 Reporting Dashboard

Module Description

Provides visibility into operational metrics.

Requirement ID	Functional Requirement
FR-028	The system shall display dashboards for attendance, fees, and academic performance.
FR-029	The system shall generate attendance reports.
FR-030	The system shall generate fee reports.
FR-031	The system shall generate result performance reports.
FR-032	The system shall allow reports to be exported to PDF and Excel.

8. Non-Functional Requirements

ID	Requirement
NFR-001	System response time shall be less than 3 seconds for standard transactions.
NFR-002	System shall support at least 1000 concurrent users
NFR-003	System shall be accessible via desktop, tablet, and mobile devices.
NFR-004	System availability shall be 99.5% during school operating hours.
NFR-005	Daily automated backups shall be performed.
NFR-006	Data shall be encrypted in transit and at rest.

9. Business Rules

BR-001	Each student must have a unique Student ID.
BR-002	A student can be assigned to only one active class per academic session.
BR-003	Attendance can only be submitted once per class per day.
BR-004	Bank transfer payments must be verified before posting.
BR-005	Receipts are generated only after successful payment posting.
BR-006	Grades are calculated automatically using predefined grading rules.
BR-007	Results must be approved before publication.
BR-008	Parents can view only information related to their own children.

10. Assumptions

- School staff will be trained before go-live.
- Parent contact information is maintained accurately.
- SMS/email services are available for notifications.
- Grading rules are approved by school management.

11. Constraints

- Must comply with school data privacy policies.
- Must support major web browsers.
- Project timeline is limited to the agreed implementation schedule.
- Budget covers only in-scope modules.
- Reliable internet access is required.

12. Acceptance Criteria

The system will be accepted when:

- All functional requirements are implemented.
- User Acceptance Testing (UAT) is completed successfully.
- Critical defects are resolved.
- User training is completed.
- Production deployment is successful.

13. Future Enhancements

- Mobile application
- Online learning integration
- Payroll management
- Timetable Management
- Transport management

14. Appendices

Appendix A – Glossary

Term Definition

SMS School Management System

UAT User Acceptance Testing